

Technology Trends Analysis Report

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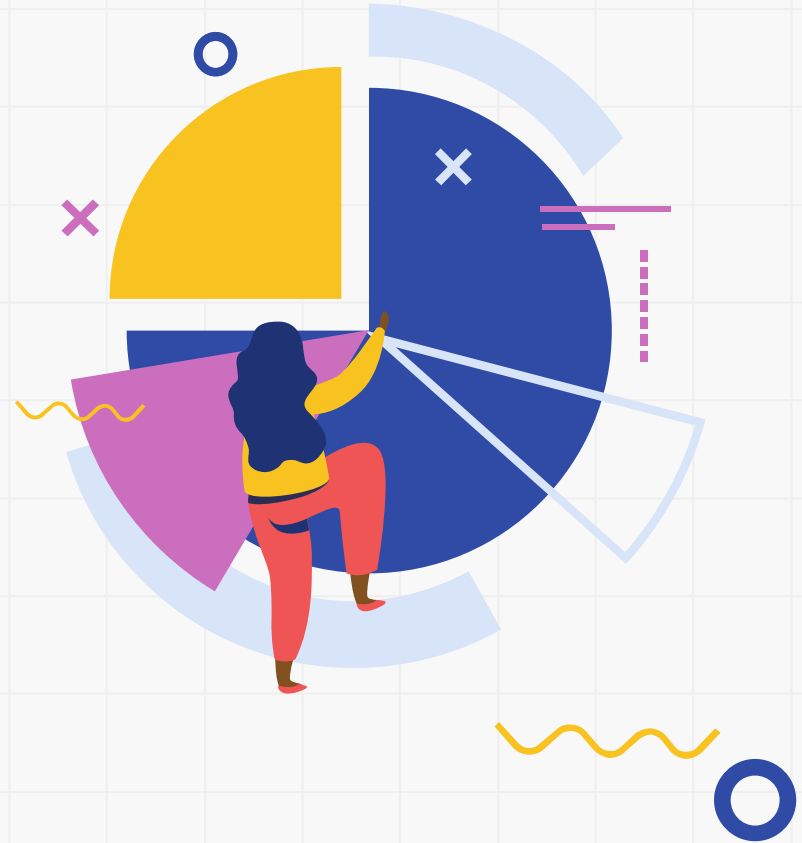




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Executive Summary

01

This project analyzes current and future technology trends using survey data. Dashboards were built in IBM Cognos Analytics to visualize programming languages, databases, platforms, web frameworks, and demographics to support data-driven insights



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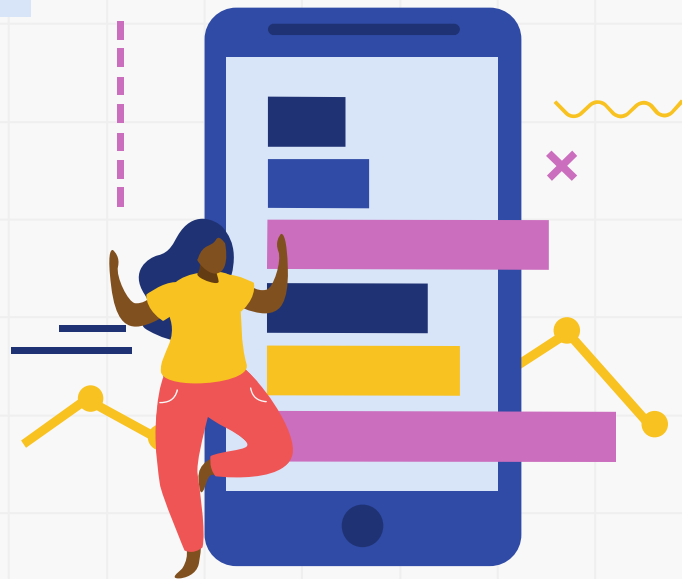
Introduction

- **Purpose:** Analyze technology usage and future trends
- **Audience:** Aspiring data professionals and decision-makers.
- **Value:** Provides insights into in-demand technologies and workforce demographics.



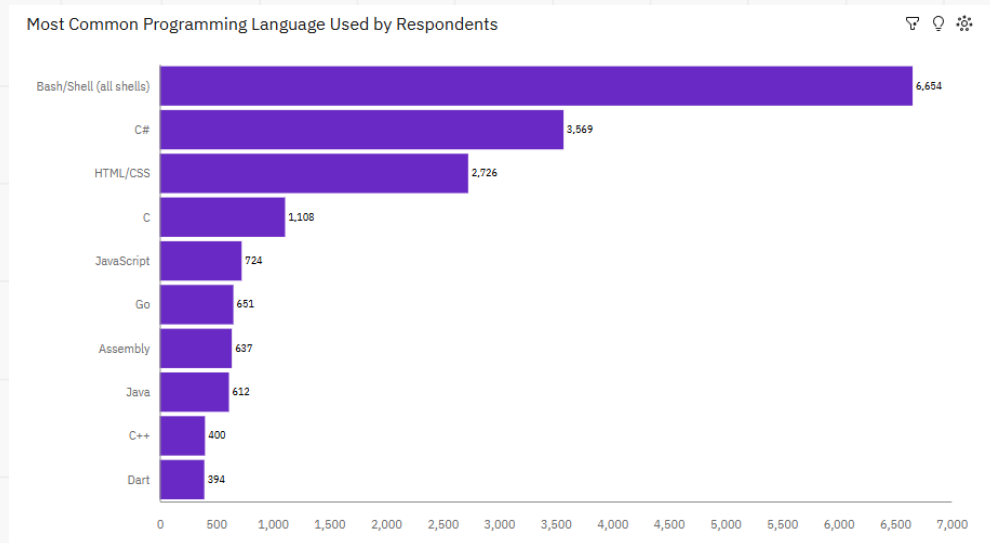
Methodology 03

- **Data Source:** survey_data_updated.csv
- **Collection:** Developer survey responses
- **Wrangling:** Removed duplicates, handled missing values, split multi-value fields using delimiter(;).
- **Tool Used:** IBM Cognos Analytics



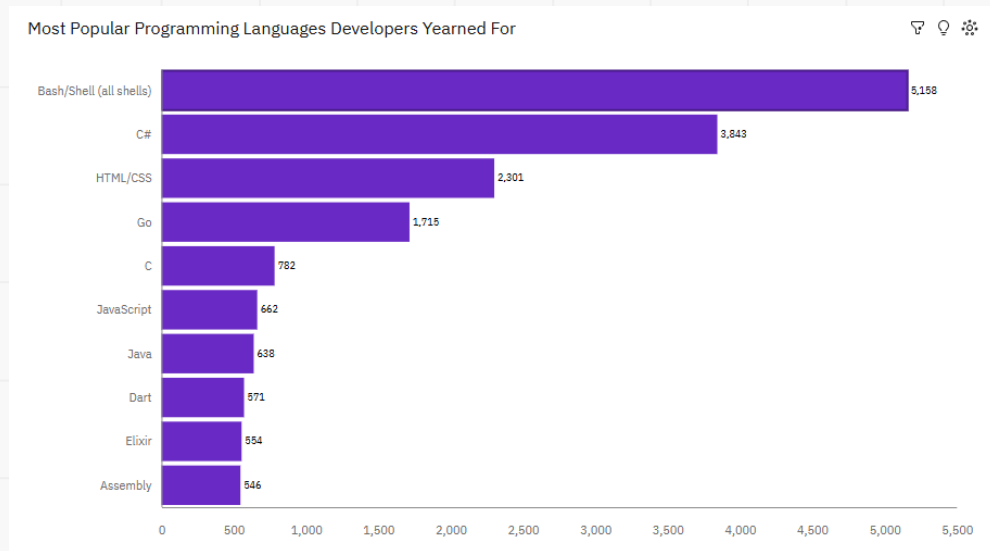
Programming Language – Current Usage

- Bar chart shows the Top 10 programming languages currently used by respondents.
- Key Insight: Bash/Shell, C#, and HTML/CSS dominate current usage.



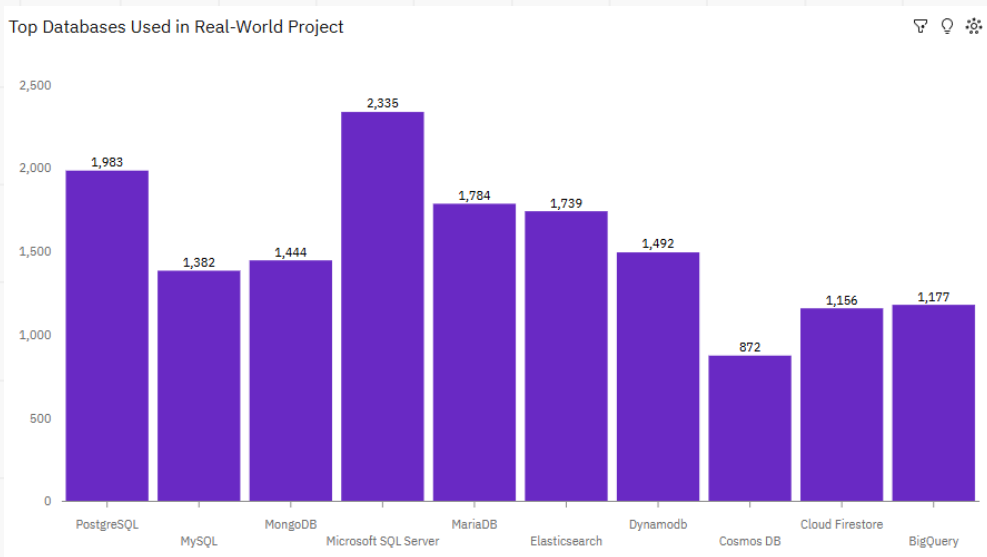
Programming Language – Future Trend

- Visualization highlights programming languages developers want to learn or use next.
- Implication: Demand is shifting toward modern and versatile languages.



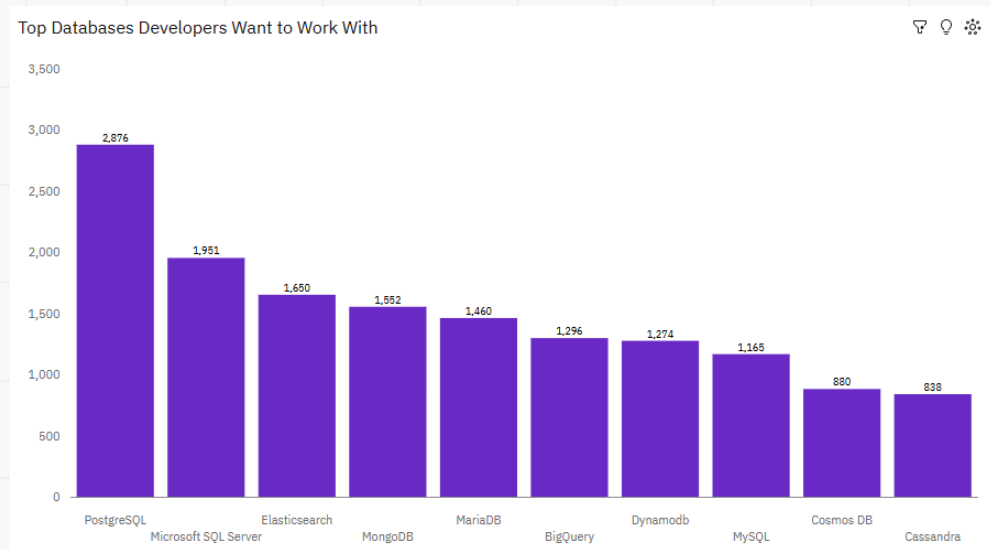
Database Trends – Current Usage

- Bar chart summarizes the Top 10 databases used in real-world projects.
- Insight: SQL-based databases remain highly relevant.



Database Trends – Future Demand

- Visualization shows databases developers want to work with in the future.
- Implication: Cloud-native and scalable databases are gaining popularity.



Dashboard Overview

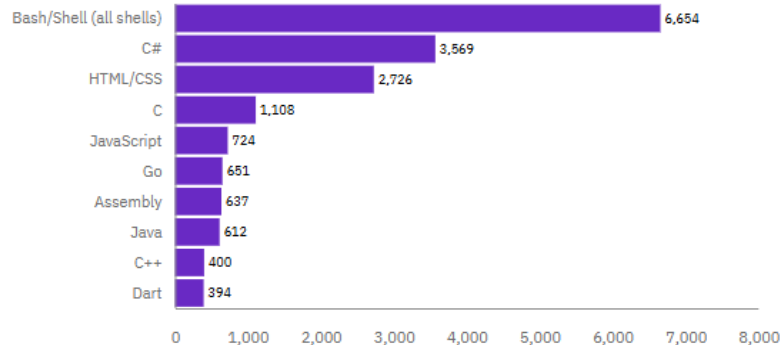
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- Three interactive dashboards were created using IBM Cognos Analytics:
 - Current Technology Usage
 - Future Technology Trends
 - Demographics

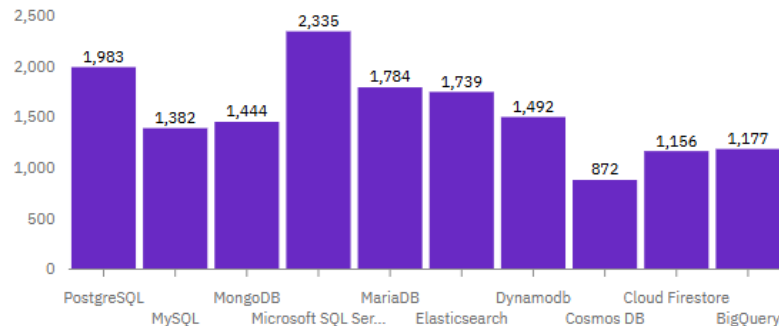


Dashboard: Current Technology Usage

Most Common Programming Language Used by Respondents



Top Databases Used in Real-World Project



Most Used Development Platforms

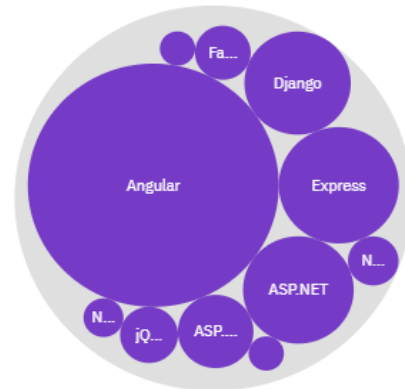


Popular Web Frameworks Among Developers



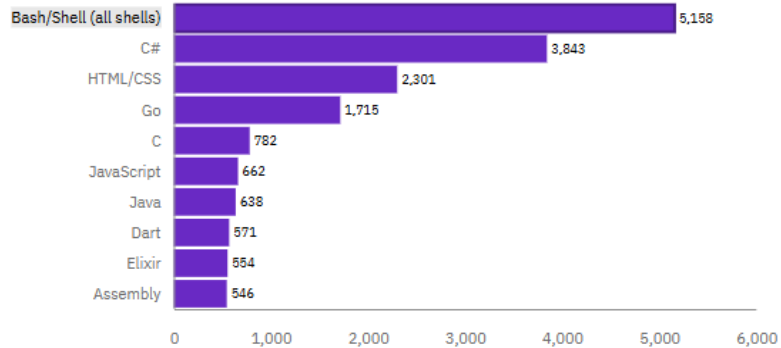
Amazon Web Services (AWS)

Microsoft Azure
Cloudflare
Google Cloud
Digital Ocean

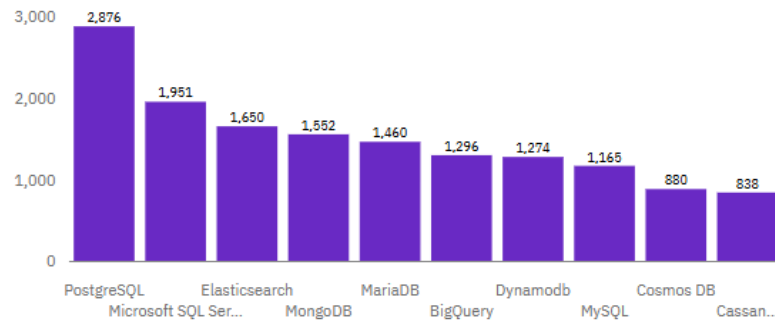


Dashboard: Future Technology Trends

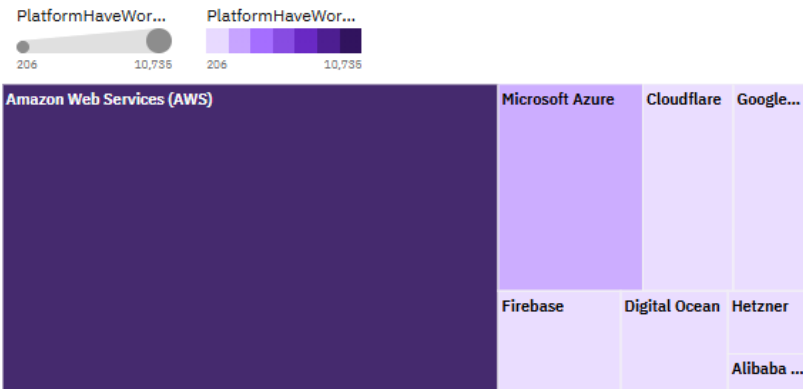
Most Popular Programming Languages Developers Yearned For



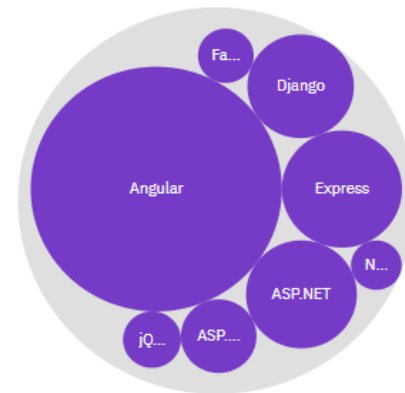
Top Databases Developers Want to Work With



Future Platform Technology Trends Among Developers



Web Frameworks Developers Want to Use

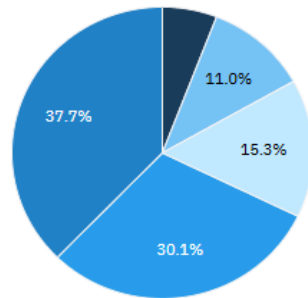


Dashboard: Demographics

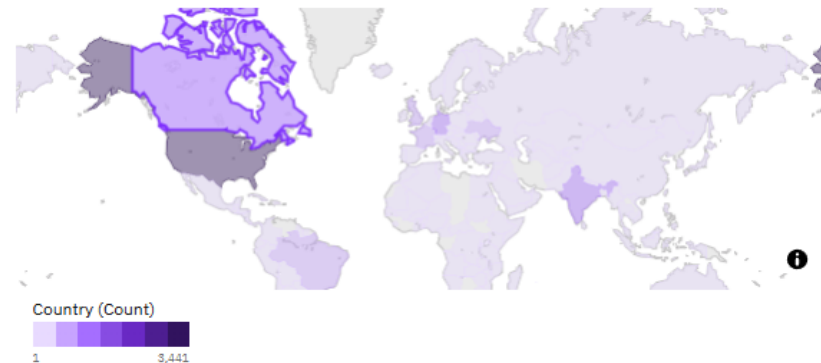
Respondent Age Distribution

Age

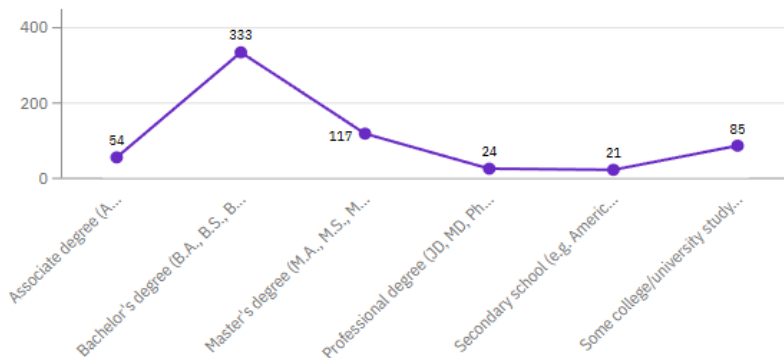
- 55-64 years old
- 18-24 years old
- 45-54 years old
- 35-44 years old
- 25-34 years old



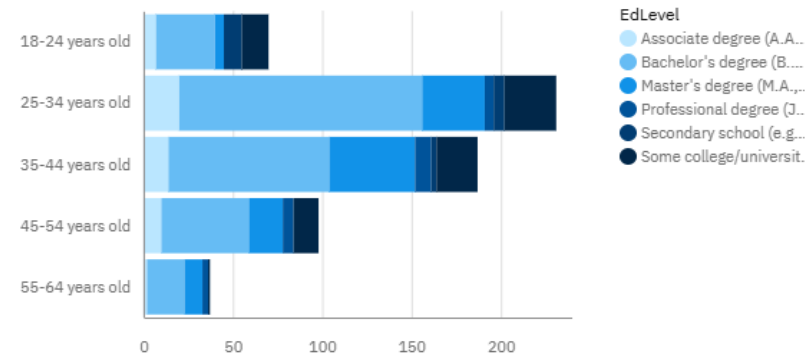
Geographic Distribution of Survey Respondents



Respondents by Formal Education Level



Age Distribution by Education Level



07

Insights from Dashboards

- AWS dominates platform usage.
- Angular and Express lead web framework interest.
- Most respondents are aged 25–44 with bachelor's degrees.





Overall Findings and Implications

- Technology trends indicate strong demand for cloud platforms and modern frameworks.
- These insights can guide learning paths and workforce planning.



Conclusion 08

- The project successfully demonstrates how dashboards can transform raw data into actionable insights.
- IBM Cognos Analytics proved effective for professional data visualization.

